

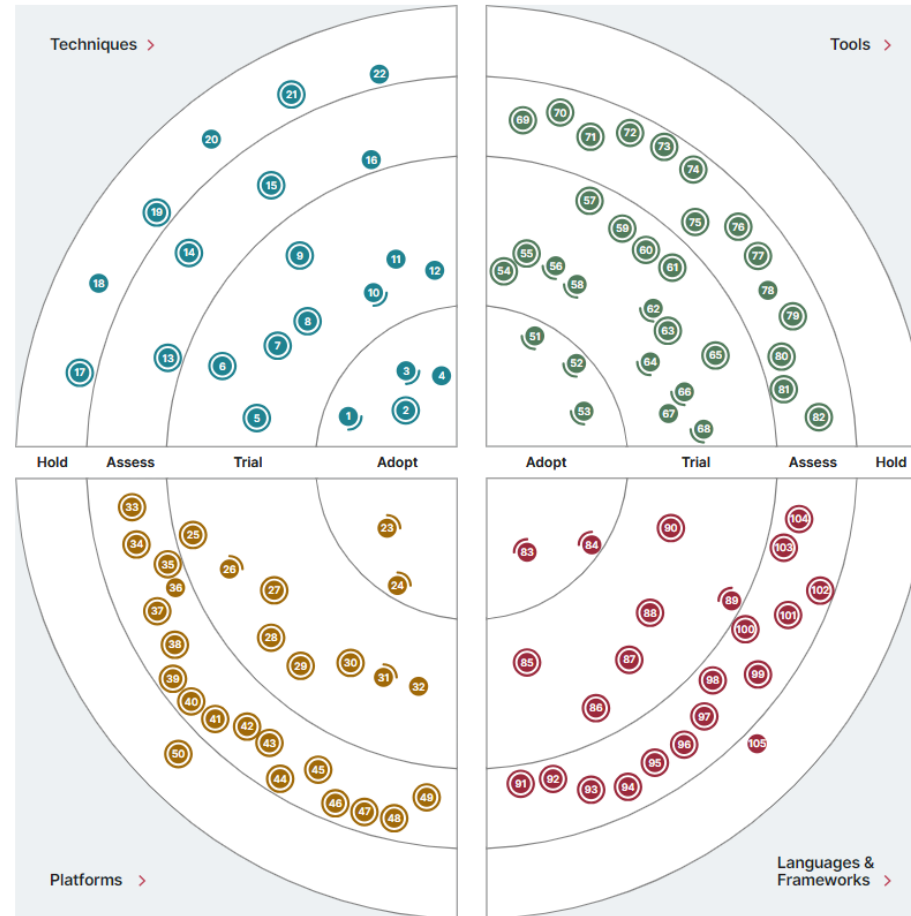
OpenTelemetry 101

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Infominds Ag



Thoughtworks Technology Radar



Technology Radar Vol. 32

Adopt ?

83. OpenTelemetry

OpenTelemetry [↗](#) is quickly becoming the industry standard for observability. The release of the [OpenTelemetry protocol \(OTLP\)](#) [↗](#) specification established a standardized way to handle traces, metrics and logs, reducing the need for multiple integrations or major rewrites as monitoring distributed solutions and interoperability requirements grow. As OpenTelemetry expands to support logs and profiling, OTLP ensures a consistent transport format across all telemetry data, simplifying instrumentation and making full-stack observability more accessible and scalable for microservices architectures.

Adopted by vendors like [Datadog](#) [↗](#), [New Relic](#) [↗](#) and [Grafana](#) [↗](#), OTLP enables organizations to build flexible, vendor-agnostic observability stacks without being locked into proprietary solutions. It supports gzip and zstd compression, reducing telemetry data size and lowering bandwidth usage — a key advantage for environments handling high volumes of telemetry data. Designed for long-term growth, OTLP ensures OpenTelemetry remains a robust and future-proof standard, solidifying its position as the de-facto choice for telemetry transport.

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What is observability

Observability refers to the ability
to **monitor, measure, and**
understand the state of an
application by examining its
outputs, logs, and metrics.

Signals

Logs

Metrics

Traces

Logs
Metrics
Traces

Logs
Metrics
Traces

Why was
OTEL created

Rise of Microservices and Kubernetes

Tools Fragmentation

Vendor lock-in

Lack of unified standards

What is OTEL

Vendor-agnostic instrumentation

Unified telemetry

Industry standard

Language	Traces	Metrics	Logs
C++	Stable	Stable	Stable
C#/.NET	Stable	Stable	Stable
Erlang/Elixir	Stable	Development	Development
Go	Stable	Stable	Beta
Java	Stable	Stable	Stable
JavaScript	Stable	Stable	Development
PHP	Stable	Stable	Stable
Python	Stable	Stable	Development
Ruby	Stable	Development	Development
Rust	Beta	Beta	Beta
Swift	Stable	Development	Development

Otel Collector

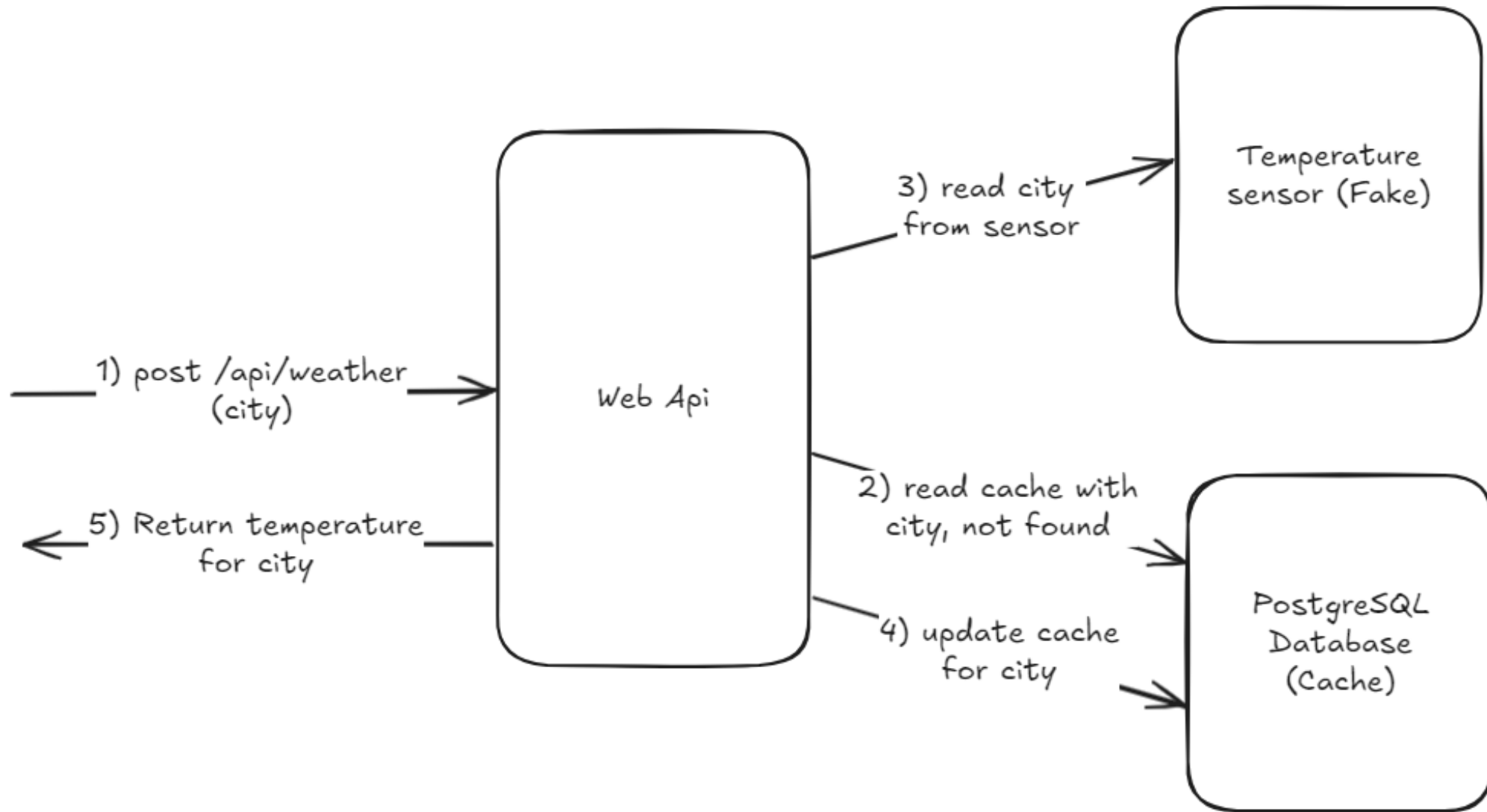
What OTEL is not

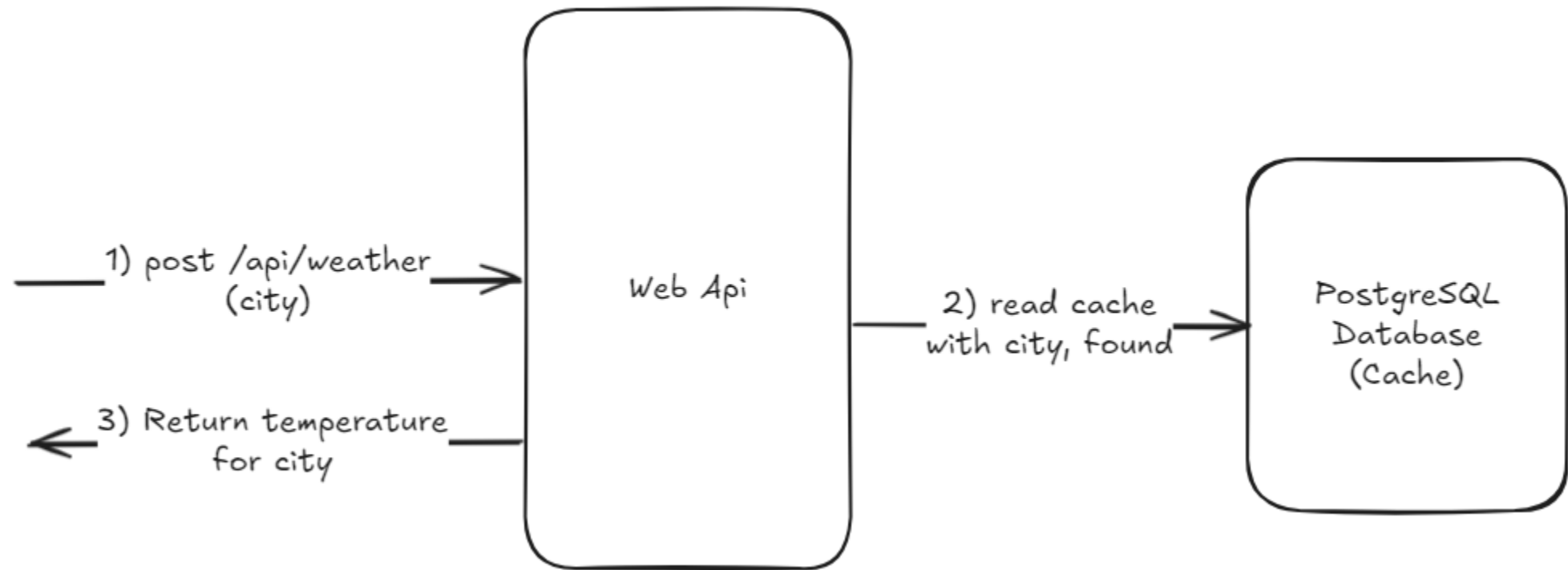
Observability Backend

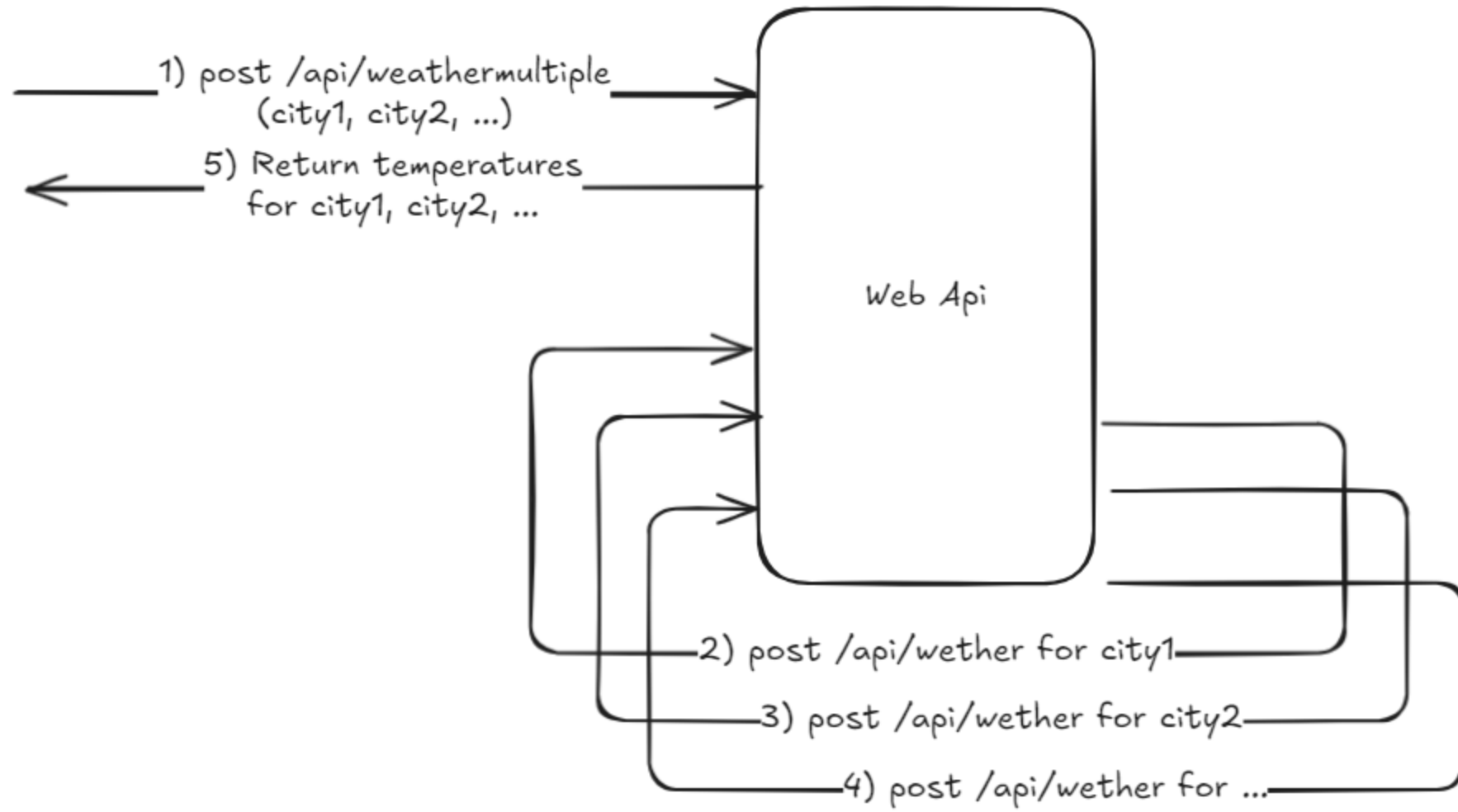
Magic Wand

Find the balance

Demo project







Aspire Dashboard (local)

Honeycomb.io (production)

Summary

Try it yourself

Links

- [Example Repo](#)
- [Open Telemetry documentation](#)
- [Aspire Dashboard](#)
- [Honeycomb](#)
- [Practical OTEL with .net 8 \(NDC\)](#)
- [OpenTelemetry in a Brownfield World \(NDC\)](#)